

Vineet kumar
Aspiring AI Engineer | AI/ML Enthusiast
(+91)7082233139 | vineetdhiran092005@gmail.com | [linkedin](#)

EDUCATION

Panipat Institute of Engineering and Technology, Samalkha (2022-2026)
Bachelor of Technology in Computer Science, Major in **AI&ML**
DAV Centenary Public School Panipat
Class XII - CBSE 2022 \ (2022)
Class X - CBSE 2020 (2020)

TECHNICAL SKILLS

Category	Skills
Languages	Python, C, C++, SQL, HTML, CSS
Libraries/Frameworks	NumPy, Pandas, Matplotlib, Seaborn, PyTorch, TensorFlow
Machine Learning and AI	Scikit-learn, NLP, FastAPI , RAG
Deep Architecture	BERT , DistilRoBERTa , HuggingFace Transformers
Tools	Git, Postman, GitHub, ChatGPT, MS Office
Databases	MySQL
Data Analysis & Visualization	Power BI, MS Excel, EDA, Statistics, Data Modeling, Data Cleaning & Preprocessing.

PROJECTS

Sentiment Aware Recommendation System

- Architected a 6-model hybrid AI pipeline combining **TF-IDF**, **BERT**, **DistilRoBERTa**, **SentenceTransformer (MiniLM-L6-v2)**, **Zero-Shot Classification (BART-MNLI)**, and Logistic Regression to deliver sentiment-aware, emotion-driven smartphone recommendations with 92–95%
- Engineered a multi-layer intent detection system using **RapidFuzz** spell correction, semantic similarity, and zero-shot reasoning to filter 95% of invalid queries.
- Designed a weighted hybrid recommendation scoring engine incorporating rating normalization, sentiment score, emotion confidence (DistilRoBERTa — 96% accuracy)

AI carrier skill gap intelligence Platform

- Developed an AI-powered career intelligence platform that parses resumes using NLP, extracts structured skills, and predicts job roles automatically using **TF-IDF + Naive Bayes** classification and **PostgreSQL**
- Engineered a 3-layer hybrid skill gap analysis engine combining exact matching, **FAISS**-powered semantic similarity using Sentence Transformer, and rule-based domain logic to accurately detect missing skills.

Heart Disease survival prediction model

- Built Designed a heart disease prediction pipeline on 897 patients combining SMOTE+ENN hybrid sampling, cost-sensitive XGBoost & Random Forest, and ensemble voting (**XGBoost + LightGBM + Random Forest**), achieving ROC-AUC of 80–85% and 72–80% sensitivity
- Learned to select algorithms and fine-tune hyperparameters to improve prediction accuracy and reduce noise
- Engineered an end-to-end ML preprocessing pipeline with imputation, log transformation, z-score normalization and SelectKBest feature selection, validated across multiple seeds for clinical robustness.

CERTIFICATIONS AND ACHIEVEMENTS

- Machine learning and python **intern – EI systems technologies**
- Development Intern - Okie dokie solutions**
- Hackathon participant – **HackOps , Savisjar (2024)**